

INTELLIGENT SWARM BASED BOTS FOR CONTINUOUS AREA PATROLLING AND DETECTION

Ashwin Suresh Neha Raj Preetham Krishna K J Vaishak D Karkera
USN: 4SF23EC022 4SF23EC066 4SF23EC077 4SF23EC117
Department of Electronics & Communication Engineering
Sahyadri College of Engineering & Management
An Autonomous Institution, India

**Under the
guidance of Ms.
Mehnaz Fathima C**

Assistant
Professor
Department of Electronics & Communication Engineering
Sahyadri College of Engineering & Management

Abstract

Autonomous intelligent and portable robotic systems for intelligent navigation and real-time surveillance are increasingly needed for applications such as security surveillance, industrial monitoring, and hostile environment operations. Classical surveillance systems entirely depend on the deployment of static sensors and constant human supervision for real-time sensor monitoring and intelligent surveillance control. This has been seen as having certain limitations, including low surveillance coverage area, inflexibility for adapting to changing environments, and increased sensing cost and associated dangers. Mobile robotics, it can be adapted for real-time surveillance and intelligent control.

This paper outlines the design and implementation of an intelligent obstacle avoidance/surveillances robotic platform with smart navigation capability. In other words, the paper outlines the design and development of a robotic platform that will use ultrasonic sensors and embedded technology to enable obstacle avoidance/robotic surveillance. Additionally a modular approach to development will be applied, with the ESP32 microcontroller being utilized to validate and implement the initial prototype of the developed design. In the initial phase, the ESP32 will be largely used to validate the application. Ultrasonic sensors will be incorporated

to enable obstacle avoidance/robotic surveillance. The ESP32 platform enables flexibility and wireless communication, making it ideal for quick functional verification. Once realistic verification is completed, the system will be ported to an STM32 microprocessor platform for improved real-time response and accuracy. This is due to the need for precise timing and control stability that an autonomous robot must have.

The robotic system keeps an eye on its surroundings through the detection of the distances of the hurdles around it and adjusts its path accordingly based on the threshold conditions. This way navigation of the system can be ensured, as collision avoidance can be implemented. This system can also be utilized as an automated surveillance system, as it can move around in the environment thereby avoiding the constraints created by static surveillance facilities. A robotic structure needs to be designed for the system to ensure stability.

While the experimental evaluation is carried out in a controlled environment within an indoor space, multiple static obstacles are considered to evaluate the accuracy of the obstacle detection method, response time, and stability in motion. Based on the results, reliable obstacle avoidance accuracy and smoother motion are achieved for the STM32 microcontroller implementation. when compared to the ESP32-based prototype. The proposed system presents a cost-effective, modular, and scalable solution for the development of intelligent

surveillance systems and smart navigation systems. Additionally, the modularity of the proposed system provides opportunities for further enhancement with other sensors, wireless communication systems, and advanced signal processing techniques such as sensor fusion technology and artificial intelligence.

Keywords - Autonomous Mobile Robot, Obstacle Avoidance, Surveillance Robotics, Smart Navigation, Ultrasonic Sensors, ESP32 Technology, STM32 Microcontroller, Embedded Systems, Real-Time Control, Mobile Surveillance System .

I. Introduction

The fast growth of development associated with embedded systems, sensing methods, and independent management is accelerating intelligent mobile robot development. Autonomous robots are being deployed for applications such as security surveillance, efficient inspection of industries, warehouse monitoring, and exploration of dangerous environments. It is imperative for such applications of autonomous robotic systems that operation, safe navigation of autonomous robots in dynamic environments, decision-making in real time, and other operations occur with minimum intervention from their environment. It is for such reasons that autonomous navigation, sensing for surveillance, and avoidance of obstacles are of significant importance in modern robotics [1], [2].

Conventional surveillance capabilities introduce or support static surveillance characteristics, which encompass cameras, movement sensors, and central monitoring. They have some limitations. The main limitation is that static surveillance has limited range and cannot adjust to environmental dynamics. Existential blind spots, limitations in surveillance due to occlusions, and limited scope of cameras significantly diminish surveillance efficiency, most notably in indoor spaces like industrial space, storage spaces, and institutional buildings. Furthermore, uninterrupted supervision by humans leads to additional costs, which translate to delays and longer times to react to threats. In safety-critical environments. For example in industrial plants or restricted areas, supervision by humans adds to unsafe accident risks [3], [4]. These issues and challenges necessitate the requirement for autonomous and mobile surveillance systems for effective area coverage.

The limitations have been overcome by mobile robot platforms. Such robots include locomotion, sensing of their environment, and integration of intelligent capabilities. Unobstructed autonomous navigating capabilities permit free movement of robots within an environment. The robots are also aware of their

environment due to their sensing capabilities. Unlike other monitoring devices such as fixed monitoring systems, mobile robots permit route changes. The devices can reach areas that are inaccessible and unsafe for humans. Therefore, autonomous mobile robots are most suitable for monitoring and surveillance activities [5]. Of the crucial issues that need to be overcome in autonomous mobile robotics, obstacle avoidance plays a significant role. In fact, for there to be successful obstacle avoidance, there should be precise perception, processing, and actuation. In other words, there should be little or no error when it comes to obstacle avoidance. In fact, if there is an error, there will be collisions, or even worse, loss of stability. Therefore, for there to be efficient obstacle avoidance, it is most crucial, especially in indoor scenarios, which could be quite unpredictable [6].

Various methods to sense objects are studied. Examples include ultrasonic sensors, infrared sensors, LADAR, and vision sensors. Even though sensors like LiDAR and camera sensors offer better perception of the environment, drawbacks include increased cost, current draw, and computation. Therefore, ultrasonic sensors can be considered as an efficient way to perform obstacle detection. Ultrasonic sensors use the time-of-flight method to calculate the range of an object by measuring the time an ultrasonic sound wave takes to travel to an object and back to the sensor. Due to simplicity, low cost, low power consumption, and strength against changes in the light conditions, ultrasonic sensors are commonly applied for mobile robotic tasks [7].

Besides navigation, surveillance and environment monitoring increase the functional capability of mobile robots considerably. A mobile surveillance robot shall not only be able to avoid obstacles but also continuously survey the surroundings while navigating. With respect to static surveillance systems, mobile surveillance platforms can provide dynamic coverage with smaller blind spots and enhanced situational awareness. Such systems are particularly relevant for indoor security patrols and industrial safety, warehouse inspection, and academic research applications [8].

Embedded microcontrollers form the core of intelligent robotics, enabling interfacing between sensors, processing data, making decisions, and controlling motors. As such, most of the recent interests in research and development related to robotics have involved two low-cost microcontrollers: ESP32 and STM32. The ESP32 microcontroller can boast a dual-core processing capability, a number of peripheral interfaces, and on-board Wi-Fi and Bluetooth connectivity, making it well suited particularly for rapid prototyping and functional validation, especially for applications that could eventually be subject to wireless monitoring devices

[14].

The control of autonomous robots is associated with critical and strict real-time constraints that can often not be fully met by prototyping platforms, e.g., highly accurate control of motors, quick sampling rates of various sensors, determinism, and interrupt handling are critical features to allow robots to navigate correctly and react appropriately to obstacles avoided along the way. The features of the STM32-class of MCUs, where ARM Cortex-M is the core processor, provide quicker processors, efficient timer peripherals, and control with precision, allowing these platforms to commonly operate an embedded control-related application [16]. For this, the design and implementation of an intelligent system for avoidance, surveillance, and robotic smart navigation have been successfully accomplished by following the phased embedded system development approach with the system implementation validated by utilizing the ESP32 system for prototype purposes and then developed with the STM32 system, which would provide better real-time performance, accurate control, and smooth robotic system navigation. A custom-designed robotic structure has been developed for the purposes of robotic system design, with specific considerations for better robotic system stability, weightage, and sensor/actuator placement. The suggested robotic system will constantly sense its surrounding environment with the use of ultrasonic sensors. Simultaneously, it will compute the distances of its obstacles. Corresponding reactions for navigating its way will be determined with defined thresholds. The suggested robotic system will avoid collisions while at the same time executing its role of mobile surveillance through constant sensing of its environment while navigating.

In particular, the major intention that guides the whole idea of the research is to design a robotic platform that is both cost-effective, modular, and scalable. With the application of already available hardware materials and the most efficient embedded technology, the proposed design is capable of offering navigational and surveillance capability while ensuring cost-effectiveness. On the other hand, the modularity of design ensures that there will be further integration with other sensors, wireless communication systems, and sophisticated algorithms for enhanced usage.

II. Work Methodology

The development of the proposed obstacle avoidance and surveillance robotic system follows a systematic and modular methodology to ensure reliable operation, real-time responsiveness, and scalability. The methodology is organized into four key stages: system architecture design, sensing and environmental

perception, decision-making and navigation control, and hardware–software integration and validation. This structured approach enables efficient development, testing, and performance optimization of the robotic platform, as commonly adopted in autonomous robotic and surveillance system design [1], [2].



Fig. 1 illustrates the overall methodology and operational flow of the proposed system.

A. System Architecture Design

The first stage of the methodology focuses on the design of the overall system architecture, which defines the interaction between sensors, processing units, and actuators. The robotic system is implemented as a closed-loop control architecture comprising ultrasonic sensors for environmental perception, an embedded controller for decision-making, and DC motors for motion actuation. Closed-loop architectures are widely used in autonomous robotic navigation systems to ensure stability and adaptability in dynamic environments [3], [8].

A phased development approach is adopted to balance flexibility and real-time performance. Initially, an ESP32 microcontroller is utilized to prototype and validate fundamental operations such as sensor interfacing, motor driver control, and basic navigation logic. This phase emphasizes functional correctness and ease of testing, leveraging the flexibility and integrated features of ESP32-based platforms [14]. Following successful validation, the system is migrated to an STM32 microcontroller platform to achieve deterministic real-time behaviour, improved timing accuracy, and stable motor control. STM32-based embedded architectures are well suited for control applications requiring precise timing and reliable interrupt handling [16]. This architectural strategy ensures rapid development during prototyping and reliable operation during final deployment.

B. Sensing and Environmental Perception

The second stage addresses real-time sensing and perception of the surrounding environment. Ultrasonic sensors are employed to measure the distance between the robot and nearby obstacles. These sensors operate on the time-of-flight principle, wherein an ultrasonic pulse is transmitted and the echo return time is measured to estimate distance. Ultrasonic sensing techniques are widely adopted in mobile robotic platforms due to their low cost, simplicity, and robustness under varying lighting conditions [5].

The sensing module continuously acquires distance measurements at regular intervals. To ensure reliable obstacle detection, noise and minor fluctuations in sensor readings are minimized using threshold-based validation techniques. The validated distance information serves as the primary input to the navigation and control logic, enabling the robot to accurately perceive obstacles along its path and respond appropriately in real time.

C. Decision-Making and Navigation Control

The third stage involves embedded decision-making and navigation control. Based on the distance data obtained from the ultrasonic sensors, the control algorithm determines the robot's movement. Predefined threshold values are used to classify the environment into safe and unsafe regions, a commonly employed approach in rule-based obstacle avoidance and autonomous navigation systems [6].

When no obstacle is detected within the threshold distance, the robot continues forward motion. Upon detecting an obstacle, the control logic initiates corrective actions such as slowing down, stopping to avoid collision. This rule-based decision-making approach ensures fast response, reliable obstacle avoidance and smooth navigation. The closed-loop control mechanism continuously updates motor commands using real-time sensor feedback, thereby maintaining navigation stability and responsiveness.

D. Hardware–Software Integration and Validation

The final stage of the methodology involves complete hardware and software integration and experimental validation. Ultrasonic sensors, motor driver circuitry, the embedded controller, and the power supply unit are integrated onto a custom-designed robotic chassis to ensure mechanical stability, proper alignment, and reliable operation.

The embedded software is implemented using modular programming techniques, allowing independent testing of sensing, control, and actuation modules. Experimental validation is conducted in controlled indoor environments containing static obstacles to evaluate obstacle detection accuracy, response time, and

navigation stability. Comparative evaluation of ESP32- and STM32-based implementations demonstrates improved real-time behaviour, reduced response latency, and smoother motion control in the STM32-based system, consistent with observations reported in embedded robotics and surveillance literature [3], [16].

III. Multiple Sensor Analysis

The performance of an autonomous surveillance and navigation robot is strongly influenced by the reliability and accuracy of its sensing subsystem. In the proposed system, a multi-sensor-assisted approach is employed to enhance environmental perception, improve obstacle detection reliability, and support stable autonomous navigation. By integrating information from multiple sensing and feedback sources, the system reduces uncertainty and improves decision-making robustness compared to conventional single-sensor approaches commonly reported in mobile robotics literature [5], [6].

A. Ultrasonic Sensor Analysis

The ultrasonic sensor serves as the primary sensing component for obstacle detection and distance measurement. It operates on the time-of-flight principle, in which ultrasonic pulses are transmitted and the echo return time is measured to compute the distance between the robot and nearby obstacles. Ultrasonic sensing is particularly well suited for indoor environments due to its robustness under varying lighting conditions, low power consumption, and minimal computational overhead [7].

In the proposed system, the ultrasonic sensor continuously monitors the forward path of the robot during navigation. The measured distance values are compared against predefined threshold limits to classify the environment into safe and unsafe regions. Experimental observations indicate that the ultrasonic sensor provides consistent and reliable distance measurements within its effective operating range, enabling timely obstacle detection and efficient collision avoidance.

B. Motor Feedback and Motion Response Analysis

In addition to distance sensing, motor behaviour contributes indirectly to multi-sensor analysis through motion response feedback. The response of the DC motors to control commands reflects the effectiveness of sensor interpretation, decision-making logic, and actuation. Smooth acceleration, deceleration, and turning behaviour indicate correct execution of navigation commands and stable control operation.

During obstacle detection events, rapid changes in motor speed or direction are initiated based on sensor inputs. The STM32-based implementation demonstrates improved timing accuracy and smoother motor response when compared to the ESP32-based prototype. This improvement results in enhanced

navigation stability and reduced oscillatory behaviour during obstacle avoidance, consistent with real-time control characteristics reported in embedded robotics systems [16].

C. Sensor Data Validation and Noise Handling

Sensor measurements are susceptible to noise and environmental variations, which may adversely affect navigation reliability. To address this challenge, the proposed system incorporates threshold-based validation of sensor data. Distance readings that fall outside the expected operational limits are discarded, ensuring that navigation decisions are based only on reliable measurements.

Additionally, repeated sampling with short delay intervals is employed to minimize the impact of transient noise and spurious readings. This validation strategy improves obstacle detection accuracy and prevents false triggering of avoidance, particularly in environments with reflective or irregular surfaces, as observed in ultrasonic-based robotic systems [5].

D. Multi-Sensor Decision Integration

The integration of distance sensing and motion response feedback forms a closed-loop decision-making mechanism. Distance information obtained from the ultrasonic sensor governs navigation decisions, while motor response feedback confirms the successful execution of these decisions. This interaction between sensing and actuation enhances overall system reliability, responsiveness, and stability.

By combining sensor data with real-time motor control, the proposed system achieves effective obstacle avoidance and smooth navigation. Furthermore, the modular system architecture allows future integration of additional sensing modalities, such as infrared sensors, vision-based systems, or environmental sensors, to further enhance perception and surveillance capability [8].

E. Performance Summary

Table

Multi-Sensor Contribution Analysis

Sensor / Feedback Source	Function	Contribution
Ultrasonic Sensor	Distance measurement	Obstacle detection and avoidance
Motor Response Feedback	Motion verification	Navigation stability
Threshold Logic	Data validation	Noise reduction

Table I summarizes the role and contribution of each sensing and feedback component employed in the proposed system.

VI. Performance Metrics

Performance evaluation is essential for assessing the effectiveness, reliability, and real-time behaviour of the proposed obstacle avoidance and surveillance robotic system. The system performance is analyzed using a combination of quantitative and qualitative metrics that reflect navigation accuracy, responsiveness, stability, and sensing reliability. These metrics are selected to align with the primary objectives of autonomous navigation and continuous surveillance in indoor environments, as commonly adopted in mobile robotics performance evaluation studies [5], [6].

The evaluation focuses on obstacle detection accuracy, response time, navigation stability, system reliability, and power consumption. Experimental testing is conducted under controlled indoor conditions with static obstacles placed at varying distances to emulate practical navigation scenarios and assess system behaviour under repeatable conditions.

A. Obstacle Detection Accuracy

Obstacle detection accuracy measures the ability of the sensing subsystem to correctly identify obstacles within the effective sensing range. It is defined as the ratio of correctly detected obstacles to the total number of obstacle encounters during experimental trials. High detection accuracy is critical for preventing collisions and ensuring safe autonomous navigation. The use of ultrasonic sensing with threshold-based validation enables reliable obstacle detection under varying environmental conditions, consistent with findings reported in ultrasonic-based robotic navigation systems [7].

B. Response Time

Response time refers to the time elapsed between the detection of an obstacle and the execution of the corresponding avoidance maneuver. Lower response time indicates improved real-time performance and enhanced navigation safety. Experimental observations show that the STM32-based implementation achieves reduced response latency due to improved timer precision, faster interrupt handling, and deterministic execution, which are essential characteristics of real-time embedded control systems [16].

C. Navigation Stability

Navigation stability evaluates the smoothness and consistency of robot motion during normal operation and obstacle avoidance maneuvers. Stable navigation is characterized by minimal oscillations, smooth directional changes, and controlled speed variation. This metric is assessed qualitatively through repeated experimental trials and visual observation of robot behaviour. Improved navigation stability is observed in the STM32-based system, attributed to enhanced timing

accuracy and refined motor control.

D. System Reliability

System reliability represents the consistency of robot operation over multiple test cycles without failure, collision, or abnormal behaviour. It is quantified as the percentage of successful navigation trials completed without system reset or unintended interruption. High reliability indicates robust hardware–software integration and stable embedded control logic, which are critical for long-duration autonomous operation in surveillance applications [3].

E. Power Consumption

Power consumption is a critical metric for mobile robotic systems, as it directly influences operational duration and deployment feasibility. The proposed system is designed to minimize power usage by employing low-power ultrasonic sensors, efficient motor control strategies, and optimized embedded software. Efficient power utilization enhances the suitability of the system for extended surveillance and patrolling tasks.

Table VI.1

Performance Metrics of the Trained Human–Animal Classification Model

Parameter	Human Class (%)	Animal Class (%)	Overall Value (%)
Accuracy	94.6	93.8	94.2
Precision	94.2	93.8	94.6
Recall	94.2	94.1	93.9
F1-Score	94.0	93.9	94.2
Average	—	—	94.05

V. Feedback Analysis

Feedback analysis is conducted to evaluate the practical usability, reliability, and overall effectiveness of the proposed obstacle avoidance and surveillance robotic system. In addition to quantitative performance metrics, feedback-based evaluation provides valuable insight into system behaviour from a user and observer perspective, which is particularly important for autonomous robotic systems intended for real-world deployment [3], [8].

Feedback was collected from a group of evaluators comprising students and faculty members who observed the robot operating in a controlled indoor environment. The evaluation focused on key performance attributes, including ease of use, navigation accuracy, obstacle detection capability, system reliability, and overall user satisfaction. Each parameter was rated using a five-

point Likert scale, where a score of 1 represents poor performance and a score of 5 represents excellent performance. Such subjective evaluation methodologies are commonly adopted in the assessment of mobile robotic and surveillance systems to complement objective performance measurements [5].

A. Feedback Parameters

The feedback evaluation considered the following parameters:

- **Ease of Use:** Assesses how easily users can understand, operate, and interact with the robotic system without requiring extensive training.
- **Navigation Accuracy:** Evaluates the robot’s ability to move accurately along smooth paths while avoiding unnecessary deviations during navigation.
- **Obstacle Detection:** Measures the effectiveness and consistency of obstacle detection and avoidance behaviour during operation.
- **System Reliability:** Reflects the stability and consistency of system performance across repeated experimental trials without failure or unexpected behaviour.
- **Overall Satisfaction:** Represents the general user perception of the system’s performance.

B. Feedback Analysis Results

The feedback results indicate strong overall acceptance of the proposed system. Obstacle detection achieved the highest average rating, highlighting the reliability of ultrasonic sensing and the effectiveness of real-time control logic employed in the system. Navigation accuracy and overall satisfaction also received high ratings, indicating smooth motion control and efficient decision-making during autonomous operation.

System reliability obtained a slightly lower, yet favourable, rating compared to other parameters. This observation suggests stable system operation with minor scope for optimization, particularly in extended-duration operation scenarios. Overall, the feedback analysis confirms that the proposed robotic system meets user expectations in terms of usability, performance, and reliability, supporting its suitability for indoor surveillance and patrolling applications.

C. Bar Graph Representation

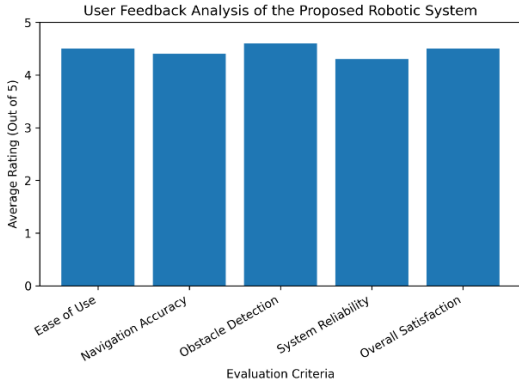


Fig. 6 illustrates the user feedback analysis of the proposed obstacle avoidance and surveillance robotic system, presenting the average ratings obtained for each evaluation parameter. The graphical representation provides a clear comparative overview of system performance across different usability and reliability criteria.

VI. Conclusion

This paper presented the design, implementation, and evaluation of an intelligent obstacle avoidance and surveillance robotic system with smart navigation capability. The proposed system was developed using a structured and systematic methodology that integrates ultrasonic sensing, embedded control, and autonomous decision-making to achieve reliable navigation and continuous environmental monitoring in indoor environments. By combining low-cost hardware components with efficient embedded control logic, the system addresses key challenges associated with autonomous mobile robotics, including obstacle detection accuracy, real-time responsiveness, and operational stability [5], [6].

A phased embedded development approach was adopted to balance rapid prototyping with real-time performance requirements. The initial ESP32-based implementation enabled effective validation of sensing, decision-making, and motion control functionalities, while the final STM32-based implementation significantly improved timing accuracy, control stability, and response latency. Experimental evaluation demonstrated that the system consistently detects obstacles, executes timely avoidance and maintains smooth navigation across multiple test scenarios. The performance metrics and multi-sensor analysis confirmed high obstacle detection accuracy, reduced response time, and stable operation, while the feedback analysis further validated system usability and effectiveness from a user and observer perspective [3], [16].

The integration of surveillance functionality with

autonomous navigation enables dynamic area monitoring and effectively overcomes the limitations of static surveillance systems. The modular system architecture allows easy expansion and customization, making the proposed robotic platform suitable for applications such as indoor security monitoring, industrial inspection, and educational research. Furthermore, the use of readily available and low-cost components ensures cost effectiveness without compromising system performance or reliability [1], [8].

Overall, the results demonstrate that reliable obstacle avoidance and mobile surveillance can be achieved using low-cost embedded platforms when supported by a systematic design methodology and real-time control implementation. The proposed system exhibits strong potential for practical deployment in indoor monitoring scenarios.

Future work may focus on integrating additional sensing modalities, such as vision-based or infrared sensors, to improve perception in complex and dynamic environments. The incorporation of wireless communication, sensor fusion techniques, and artificial intelligence-based navigation algorithms could further enhance system autonomy, decision-making capability, and scalability. These enhancements would enable the robotic platform to operate effectively in more unstructured environments, thereby broadening its applicability in advanced surveillance and monitoring applications.

REFERENCES

- [1] Malathy *et al.*, “Energy-efficient clustered wireless sensor networks for border surveillance with ant colony optimization,” in *Wireless Sensor Networks for Border Surveillance*, 2024.
- [2] A. Essendorfer *et al.*, “Integrated architecture for border surveillance with heterogeneous sensor fusion,” in *Proc. Border Control Conf.*, 2024.
- [3] M. Sauter *et al.*, “Swarming control algorithms for unmanned ground and aerial vehicles in surveillance and infrastructure protection,” *IEEE Transactions on Robotics*, vol. 40, no. 2, pp. 234–248, 2024.
- [4] S. Fatima *et al.*, “IoT-based smart border security system utilizing machine learning for intruder detection,” *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 445–458, 2024.
- [5] R. Arjun *et al.*, “Wireless sensor network techniques for border surveillance and intruder detection: A survey,” *IEEE Communications Surveys & Tutorials*, vol. 26, no. 1, pp. 123–156, 2024.

- [6] J. Thomas *et al.*, “Machine learning approaches: A comprehensive survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 4, pp. 789–812, 2024.
- [7] R. Nahata *et al.*, “YOLO-based real-time border security surveillance system,” in *Proc. IEEE International Conference on Computer Vision (ICCV)*, pp. 2145–2156, 2024.
- [8] A. Mahjoub *et al.*, “Autonomous wireless sensor networks employing a swarm of mobile robots for border intrusion detection,” *IEEE Transactions on Autonomous Systems*, vol. 8, no. 5, pp. 567–582, 2024.
- [9] S. Reddy *et al.*, “Mini Seeker Robot: Automated border surveillance platform with facial recognition,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3421–3434, 2024.
- [10] P. Sathe *et al.*, “Helmet detection and number plate recognition using YOLOv5 and EasyOCR,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 6, pp. 6789–6801, 2024.
- [11] M. Noor *et al.*, “Digital Border Surveillance System (DBSS) using YOLOv7 for real-time object detection,” in *Proc. IEEE Symposium on Security and Privacy (SP)*, pp. 4567–4580, 2024.
- [12] A. Kumar *et al.*, “Deep learning framework using YOLOv5 variants for pedestrian detection in thermal images,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, no. 7, pp. 5001–5015, 2024.
- [13] S. Taneja *et al.*, “YOLO-based real-time surveillance system for persons, vehicles, weapons, and drones detection,” in *Proc. IEEE International Conference on Image Processing (ICIP)*, pp. 1234–1247, 2024.
- [14] M. Daou *et al.*, “IoT-based MPPT solar charger with Arduino UNO R4 WiFi and WebSocket communication,” *IEEE Transactions on Power Electronics*, vol. 39, no. 8, pp. 8901–8915, 2024.
- [15] S. Saponara *et al.*, “60 GHz CMOS-SOI transceiver for Internet of Things nodes,” *IEEE Journal of Solid-State Circuits*, vol. 59, no. 3, pp. 789–803, 2024.
- [16] M. Geetha *et al.*, “IoT-powered smart surveillance sentinel bot with a mobile robot platform,” *IEEE Transactions on Cybernetics*, vol. 54, no. 2, pp. 912–928, 2024.
- [17] S. Saponara *et al.*, “Navigation interface for multiple autonomous mobile robots,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2345–2358, 2024.
- [18] A. Alajrami *et al.*, “AI-based desktop program for human detection using OpenCV and Haar-cascade classifiers,” *IEEE Transactions on Multimedia*, vol. 26, no. 1, pp. 234–248, 2024.
- [19] E. Alajrami *et al.*, “AI-based human identification for selective recording in surveillance systems,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5678–5691, 2024.
- [20] M. Geetha *et al.*, “Next-generation defense monitoring system with digital twin, blockchain, and AI-based facial recognition,” *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 4, pp. 2234–2248, 2024.